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## Global Macro Commentary | Global

## October 22: Geopolitics in Focus

US proposes new sanctions on Russia & expanded software export controls on China; potential lower capital requirement hikes in Basel Endgame; potential US-India trade agreement; soft UK CPI; Bank Indonesia on pause despite expected 25bp cut; DXY at 98.89 (0.0%); US 10y at 3.951 (-1.2bp).

**Risk sentiment softens throughout the day amid geopolitical developments involving the US, China, and Russia.**

## Developed Markets

- Treasury swap spreads widen on the back of unconfirmed reports that the Fed is considering even smaller capital requirement increases for large banks in the upcoming Basel Endgame re-proposal.** The rumored 3-7% increase, if accurate, is slightly higher than our banks analysts' expectations (for more, see [here](#)). The afternoon's 20y re-opening stops 1.2bp through, showing strong demand. Media reporting suggests the Fed [lost access](#) to ADP's private payrolls data, further limiting data availability amid the government shutdown. Some Republican lawmakers [suggest](#) a 1-year continuing resolution to end the shutdown, but it remains unclear if Democrats would support such a plan, and the House of Representatives is still out of session.
- US yields edge down 1bp, supported by unwinding momentum and deteriorating risk sentiment as the US administration considers expanding export controls on China.** The proposed measures could apply globally to items containing or produced using US software, potentially creating more negotiating leverage for the US ahead of the potential meeting between President Trump and Chinese President Xi at the APEC Summit on October 31. Regarding the proposed measures, Treasury Secretary Bessent [states](#) that "everything is on the table." The downward correction in the S&P 500 (-0.5%) is led by AI-related stocks and other sectors exposed to trade disruptions (Industrials: -1.3%; Consumer Discretionary: -1.0%).
- Oil prices rise 2-3% on reports of declining US inventory and geopolitical developments with Russia.** The EIA reports that US crude stocks fell by nearly 1 million barrels last week. Brent prices climb further in the afternoon as the US Treasury [targets](#) two Russian oil companies with sanctions unless there is an "immediate ceasefire." However, President Trump [refutes](#) media reporting that the US has authorized Ukraine to use key military equipment. The oil-sensitive NOK (USD/NOK: -0.4%) and CAD (USD/CAD: -0.2%)

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outperform among G10 currencies.

- **The dollar trades sideways (DXY: 0%), and GBP underperforms (GBP/USD: -0.1%) after UK September CPI comes in softer than expected** at 3.8% y/y (C: 4.0%; P: 3.8%). Gilts bull-steepen sharply (2y: -9bp; 30y: -4bp) as market pricing for a BoE cut by December rises to ~70%. Our economists [believe](#) a November cut is now a possibility, but the BoE's firm focus on elevated headline inflation still suggests risks are skewed towards December. Meanwhile, euro area rates sell off in a modest bear-steepening (30y Bund: +2bp), despite some early morning spillover from the UK rates rally. EUR/USD climbs 0.1% in line with yield differentials.
- **Japanese rates rally ~1bp across the curve, with the long end showing little reaction to [reports](#) of Prime Minister Takaichi preparing a \$92bn economic stimulus package.** Expectations for some fiscal expansion under the new administration appear to be largely priced by markets. In the Tokyo afternoon, short-end JGBs find support from a solid BoJ JGB purchase operation. USD/JPY closes little changed.

#### Emerging Markets

- **Indian media reports progress towards a trade agreement with the US that could reduce US tariffs to 15-16% from the current 50%.** According to unconfirmed reports, the agreement could be announced at the ASEAN Leaders Summit on October 26-28 and involve India gradually reducing its imports of Russian oil. Amid a local Indian holiday, derivative markets imply a stronger INR, with 1m USD/INR NDF falling 0.3%.
- **AxJ currencies mostly remain stable while rates rally, but a surprise decision by BI drives a sell-off in Indonesian rates.** 5y IndoGBs sell off 4bp after BI unexpectedly maintains its policy rate at 4.75%, whereas consensus expected a 25bp cut. IDR initially weakens 0.2% but retraces to end the Asia session largely unchanged. Meanwhile, TWD weakens 0.3% as Taiwan's CBC discloses \$13bn of FX intervention in 1H25.
- **Rates also bull-flatten in Thailand (5y THB NDIRS: -5bp) after BoT lowers its 4Q25 GDP growth forecast, and the minutes from BoT's October meeting convey a dovish outlook.** THB weakens another 0.2% against USD as gold prices continue to fall (-1%) following Tuesday's [historic sell-off](#). Meanwhile, Korean rates outperform (5y KRW NDIRS: -5bp) ahead of Thursday's BoK meeting, where our economists [expect](#) the BoK to remain on hold in light of the recent Seoul housing price rebound (for more, see [here](#)).
- **LatAm currencies trade mixed, with Chile reversing weakness from Tuesday (USD/CLP: -0.3%) on stronger copper prices.** Brazil's DI curve bull-flattens and BRL weakens 0.4% as market participants await details on Finance Minister Haddad's push to reincorporate key elements of rejected fiscal measures (MP 1303) into new bills. COP underperforms, declining 0.4%

against USD despite a lack of new developments regarding the recent US-Colombia tensions. Mexico's TIE and Colombia's IBR curves trade relatively flat in line with US rates.

- **In CEEMEA, rates mostly sell off across the region in line with euro area rates.** In Turkey, rates sell off ~12bp across the OIS curve while local equities outperform (BIST 30: +0.9%) ahead of Thursday's CBT meeting, where our economists expect the CBT to lower rates by 150bp (C: -100bp). In South Africa, the entire curve shifts higher by ~6bp following several sessions of bull-flattening.

## Central Bank Monitor

### Developed Markets

- **ECB Vice President Guindos** [said](#) current interest rates are "adequate," and euro area inflation risks are balanced.
- **ECB President Lagarde** [called](#) for greater trading integration across the European Union.

### Emerging Markets

- **In Indonesia, BI** [unexpectedly](#) maintained its policy rate at 4.75% (MSe: 4.50%; C: 4.50%; P: 4.75%), citing inflation expectations and the need to maintain IDR stability. Governor Warjiyo [said](#) the central bank still sees room for lower rates given low inflation projections and below-capacity economic growth. He mentioned the transmission of recent rate cuts, growth and inflation prospects, and IDR as factors to assess the room for further easing. He called for banks to accelerate reduction in lending rates and floated up to a 500bp RRR cut for banks that lend to priority sectors, with an additional 50bp cut for banks that align their credit rate with the policy rate.
- **In Thailand, BoT** [revised](#) down its 3Q25 GDP growth forecast to -0.5% q/q while expecting a 0.5% recovery in 4Q25 as plants reopen and the fiscal stimulus boosts consumption. BoT's October [meeting minutes](#) express concerns around below-potential growth. The MPC noted financial conditions are tight and expressed concerns around the impact of USD/THB weakness on exporters. The MPC agreed that monetary policy should remain accommodative, emphasizing the importance of preserving policy space, though some members said monetary policy should also consider financial stability. The two members who voted for a 25bp cut cited economic vulnerabilities, the need for more accommodative financial conditions, and reducing the risk of low inflation from demand-side factors.
- **In Taiwan, CBC** [disclosed](#) that it bought \$13.25bn of foreign currency in 1H25, the most since 2H20. The central bank sees global trade growth weakening in 2026 due to US tariffs weighing on Taiwan's exports and private investment growth while private consumption recovers.

**Exhibit 1:** G4 Rates Closes (5pm NY)

| Tenor | US     |           |     | Germany |           |     | UK     |           |     | Japan  |           |     |
|-------|--------|-----------|-----|---------|-----------|-----|--------|-----------|-----|--------|-----------|-----|
|       | 22-Oct | 1d Δ (bp) | 2wk | 22-Oct  | 1d Δ (bp) | 2wk | 22-Oct | 1d Δ (bp) | 2wk | 22-Oct | 1d Δ (bp) | 2wk |
| 2y    | 3.445  | (1.1)     |     | 1.907   | 0.2       |     | 3.762  | (9.0)     |     | 0.919  | (1.2)     |     |
| 3y    | 3.444  | (1.4)     |     | 1.957   | 0.6       |     | 3.742  | (8.8)     |     | 1.000  | (1.5)     |     |
| 5y    | 3.553  | (1.1)     |     | 2.158   | 0.5       |     | 3.870  | (7.2)     |     | 1.207  | (1.4)     |     |
| 7y    | 3.730  | (1.3)     |     | 2.299   | 0.9       |     | 4.016  | (6.5)     |     | 1.422  | (1.9)     |     |
| 10y   | 3.949  | (1.3)     |     | 2.562   | 1.1       |     | 4.416  | (6.0)     |     | 1.639  | (1.6)     |     |
| 20y   | 4.505  | (1.2)     |     | 3.044   | 1.5       |     | 5.097  | (5.1)     |     | 2.624  | (0.7)     |     |
| 30y   | 4.531  | (1.2)     |     | 3.146   | 1.9       |     | 5.220  | (4.5)     |     | 3.103  | (1.4)     |     |

Source: Morgan Stanley Research, Bloomberg

**Exhibit 2:** Inflation Closes (5pm NY)

| Tenor | US TIPS |           |     | 10y Real Yields, BEI |        |           | 10y  | 10y Real Yields, BEI |           |     |
|-------|---------|-----------|-----|----------------------|--------|-----------|------|----------------------|-----------|-----|
|       | 22-Oct  | 1d Δ (bp) | 2wk | 10y                  | 22-Oct | 1d Δ (bp) |      | 22-Oct               | 1d Δ (bp) | 2wk |
| 5y    | 1.176   | (3.4)     |     | DBRi                 | 0.617  | 0.0       | BTPi | 1.667                | (0.2)     |     |
| BEI   | 2.353   | 2.4       |     | BEI                  | 1.737  | 1.0       |      | 1.705                | 0.7       |     |
| 10y   | 1.658   | (2.8)     |     | UKTi                 | 1.512  | (4.1)     |      |                      |           |     |
| BEI   | 2.289   | 1.4       |     | BEI                  | 2.976  | (1.8)     |      |                      |           |     |
| 30y   | 2.320   | (2.5)     |     | JGBi                 | 0.055  | (1.3)     |      |                      |           |     |
| BEI   | 2.213   | 1.2       |     | BEI                  | 1.560  | (0.3)     |      |                      |           |     |

Source: Morgan Stanley Research, Bloomberg

**Exhibit 3:** Macro Closes (5pm NY)

| Spot | Majors |          |     | Spot | EM      |          |     | Index     | Equities / Commodities |          |     |
|------|--------|----------|-----|------|---------|----------|-----|-----------|------------------------|----------|-----|
|      | 22-Oct | 1d Δ (%) | 2wk |      | 22-Oct  | 1d Δ (%) | 2wk |           | 22-Oct                 | 1d Δ (%) | 2wk |
| DXY  | 98.90  | (0.0)    |     | CNH  | 7.13    | (0.0)    |     | S&P       | 6699.40                | (0.5)    |     |
| EUR  | 1.161  | 0.1      |     | INR  | --      | --       |     | Stoxx     | 5639.21                | (0.8)    |     |
| GBP  | 1.336  | (0.1)    |     | THB  | 32.88   | 0.4      |     | FTSE      | 9515.00                | 0.9      |     |
| JPY  | 151.98 | 0.0      |     | KRW  | 1431.10 | (0.1)    |     | Nikkei    | 49307.79               | (0.0)    |     |
| CHF  | 0.796  | (0.1)    |     | ZAR  | 17.41   | 0.1      |     | VIX       | 18.60                  | 4.1      |     |
| AUD  | 0.649  | 0.0      |     | MXN  | 18.43   | (0.0)    |     | Gold (S)  | 4098.42                | (0.6)    |     |
| CAD  | 1.399  | (0.2)    |     | BRL  | 5.40    | 0.2      |     | WTI (Fut) | 58.50                  | 2.2      |     |

Source: Morgan Stanley Research, Bloomberg

**Exhibit 4:** G4 Central Bank Implied Pricing

| Central Bank OIS |         |           |         |               |         |           |         |               |         |           |         |               |         |           |         |
|------------------|---------|-----------|---------|---------------|---------|-----------|---------|---------------|---------|-----------|---------|---------------|---------|-----------|---------|
| Fed              | Implied | 1d Δ (bp) | Total Δ | ECB           | Implied | 1d Δ (bp) | Total Δ | BOE           | Implied | 1d Δ (bp) | Total Δ | BOJ           | Implied | 1d Δ (bp) | Total Δ |
| EFFR (10/21)     | 4.110   | 0.0       | —       | EONIA (10/21) | 1.928   | 0.0       | —       | SONIA (10/21) | 3.969   | 0.0       | —       | TONAR (10/21) | 0.477   | 0.0       | —       |
| 10/29/25         | 3.863   | 0.5       | (24.8)  | 10/30/25      | 1.921   | (0.1)     | (0.7)   | 11/06/25      | 3.878   | (5.9)     | (9.1)   | 10/30/25      | 0.502   | 1.8       | 2.5     |
| 12/10/25         | 3.615   | 0.3       | (49.5)  | 12/18/25      | 1.887   | 0.2       | (4.1)   | 12/18/25      | 3.790   | (7.5)     | (17.9)  | 12/19/25      | 0.568   | (7.3)     | 9.1     |
| 01/28/26         | 3.463   | 0.5       | (64.8)  | 02/05/26      | 1.865   | 0.2       | (6.3)   | 02/05/26      | 3.643   | (8.8)     | (32.6)  | 01/23/26      | 0.685   | (0.1)     | 20.8    |
| 03/18/26         | 3.328   | 0.3       | (78.2)  | 03/19/26      | 1.803   | (0.5)     | (12.5)  | 03/19/26      | 3.581   | (10.1)    | (38.8)  |               |         |           |         |
| 04/29/26         | 3.248   | (0.2)     | (86.3)  | 04/30/26      | 1.814   | 2.6       | (11.4)  | 04/30/26      | 3.476   | (11.0)    | (49.3)  |               |         |           |         |
| 06/17/26         | 3.106   | (0.4)     | (100.4) | 06/11/26      | 1.777   | 2.3       | (15.1)  | 06/18/26      | 3.439   | (10.8)    | (53.0)  |               |         |           |         |

Source: Morgan Stanley Research, Bloomberg

**The Day Ahead***Developed Markets*

- 7:45pm BST - **France October Business Confidence** (MS: 96; C: 96; P: 96)
- 3:00pm BST - **EA October Consumer Confidence** (MS: -14.6; C: -15; P: -14.9)

*Emerging Markets*

- Thu - **Korea BoK Rate Decision** (MSe: 2.50%; C: 2.50%; P: 2.50%). Our economists expect the BoK to keep rates on hold with a 5-1 vote given the recent Seoul house price rebound. [Read more](#)
- 12:00pm BST – **Turkey CBT Rate Decision** (MSe: 39.0%; C: 39.5%; P: 40.5%). Our economists expect the CBT to slow the pace of easing by delivering a 150bp rate cut at its upcoming October meeting, compared to 250bp in September. [Read more](#)

- 1:00pm HKT - **Singapore September Headline CPI** (C: 0.6% y/y; P: 0.5%) & **Core CPI** (C: 0.2% y/y; P: 0.3%)
- 4:30pm HKT - **Hong Kong September CPI** (C: 1.1%; P: 1.1%)
- 9:00am BST - **Poland September Unemployment Rate** (C: 5.6%; P: 5.5%)
- 8:00am EST - **Mexico IH October CPI** (MSe: 0.34% h/h; P: 0.02%; C: 0.38%). Headline inflation is expected to accelerate sharply from last month's near-flat bi-weekly print, driven by seasonal pressures in energy and food. We also expect core CPI to rise to 0.20%, but with a more modest increase, reinforcing our view that underlying price pressures remain contained. The market is currently pricing in ~41bps of cuts for the remainder of 2025 and 32bps for next year, with a terminal rate of 6.77% in March
- 8:00am EST - **Mexico August Retail Sales** (P: 2.40% y/y; C: 2.60%). Retail activity is expected to show a slight pickup, consistent with resilient domestic demand

## Economic Releases

### Developed Markets

- 10:00am EST - **US September Existing Home Sales** (MSe: 4.060m; C: 4.1m; P: 4.0m). After a net 3.7% increase in the pending home sales index over the past two months, we forecast a 1.5% increase in existing home sales to a 4.060 million annual rate. Existing home sales have been running close to flat for about two years (with some monthly exceptions), but rising inventories of existing homes for sale appear to be putting downward pressure on home price appreciation, which eventually should support a somewhat faster sales pace. High rates remain an important headwind.
- 11:00am EST - **US October Kansas City Fed Manufacturing Activity** (C: 2; P: 4). September's index rose to 4 from 1 in August, signaling modest expansion, though future expectations softened with the forward-looking composite index falling to 7 from 11. We forecast October's reading to hold around 4, reflecting continued but subdued growth amid persistent weakness in new orders and export demand.
- **UK September CPI: 3.8% y/y** (C: 4.0%; P: 3.8%); **Core CPI: 3.8%; Services CPI: 4.7% & RPI: 4.5%**. Our economists note underlying services inflation, on our estimates, came in at 3.9% y/y, the lowest such print since February 2022. Core goods inflation was below market expectations, with a decent miss on food despite a further sharp rise in beef prices. [Read more](#)

### Emerging Markets

- **Korea September PPI: 1.2% y/y** (P: 0.6%)
- **Malaysia September CPI: 1.5% y/y** (C: 1.4%; P: 1.3%)
- **Hungary September Unemployment Rate: 4.5%** (C: 4.5%; P: 4.4%)
- **Poland September Retail Sales: 6.6% y/y** (C: 7.2%; P: 3.0%)
- **South Africa September CPI: 3.4% y/y** (MSe: 3.5%; C: 3.4%; P: 3.3%)
- **Mexico August Economic Activity IGAE: -0.90% y/y** (P: -1.14%; C: -0.50%). Activity remains weak, but the pace of contraction eased slightly in August. It is still softer than expected, suggesting a slow recovery.

For more details, please refer to our economists' weekly reports and data previews:

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- [Japan](#)
- [Europe](#)
- [Asia/Pacific](#)
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# G4 Smarter (Beta) Trading Partners

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Since 2000, enhancements to a G4 10y government bond futures momentum strategy have produced higher Sharpe ratios and stronger returns relative to total return government bond indices for the G4: US, Germany, Japan, and the UK. See [A "Smarter" \(Beta\) Way to Trade G4 10y Futures Duration?](#) for more information on these strategies.

## Exhibit 5: Trading Signals for G4 Smarter (beta) Trading Strategy

| Current Risk, G4 10y Futures | G4 Strategy Weight | Trade Longs Portfolio | Fade Shorts Portfolio | Total Risk Trade Longs Only | Total Risk Trade Longs/Fade Shorts (max 200%) | Trade Longs Portfolio Entry Date | Trade Longs Portfolio Exit Date | Fade Shorts Portfolio Entry Date | Fade Shorts Portfolio Exit Date |
|------------------------------|--------------------|-----------------------|-----------------------|-----------------------------|-----------------------------------------------|----------------------------------|---------------------------------|----------------------------------|---------------------------------|
| J8 10y Future                | 32.50%             | 100%                  | 100%                  | 100%                        | 200%                                          | 10/15/2025                       | 11/20/2025                      | 9/19/2025                        | 10/28/2025                      |
| GE 10y Future                | 29.25%             | 100%                  | 0%                    | 100%                        | 100%                                          | 10/7/2025                        | 11/11/2025                      | -                                | -                               |
| US 10y Future                | 30.50%             | 100%                  | 100%                  | 100%                        | 200%                                          | 10/14/2025                       | 11/19/2025                      | 9/24/2025                        | 10/30/2025                      |
| UK 10y Future                | 7.75%              | 100%                  | 100%                  | 100%                        | 200%                                          | 10/15/2025                       | 11/19/2025                      | 9/25/2025                        | 10/30/2025                      |

Source: Morgan Stanley Research

### Trading Strategy 1 – "Trade Longs/Fade Shorts"

When the 5-day moving average crosses above the 20-day moving average, buy the futures contract (long duration) and hold for a 25-business-day period. When the 5-day moving average crosses below the 20-day moving average, buy the futures contract and hold for a 25-business-day period. In short, this strategy buys futures when the Simple Moving Average Crossover (SMAX) generates both a long and a short signal, given the historical outperformance of long signals traded long and the underperformance of short signals traded short. Given that the SMAX could generate both a long and a short signal within the predefined holding period, an investor may have a 200% long position since each of the two signals would be traded in separate portfolio sleeves.

### Trading Strategy 2 – Trade "Longs Only"

When the 5-day moving average crosses above the 20-day moving average, buy the futures contract (long duration) and hold for a 25-business-day period. When the 5-day moving average crosses below the 20-day moving average, do nothing. In short, an investor *only* trades long signals initiated by the SMAX given their historical precedent to outperform.

# US Holiday Trading Strategy

## Currently Active Strategies

None.

Combining the individual trading signals based on 10y Treasury future (TY) price patterns before and after US holidays has generated an annual strategy with no down years since 1987. See [Trading 10y Treasury Futures Around US Holidays](#) for more information.

Using historical data since 1983, we created a trading strategy using 10y Treasury futures (TY) that trades 11 times per year, either long or short, before or after the following US holidays: New Year's Day, Martin Luther King Day, Presidents' Day, Good Friday, Memorial Day, Independence Day, Labor Day, Columbus Day, Thanksgiving, and Christmas. [Exhibit 6](#) displays the trading rules:

**Exhibit 6:** Trading rules for US holiday trading model using US 10y futures contracts (TY)

| Holiday                     | Day of Week | Direction | Before/After Holiday | Trading Rule (business days, 3pm NY unless specified)                  | Trade Entry (business days) | Trade Exit (business days) |
|-----------------------------|-------------|-----------|----------------------|------------------------------------------------------------------------|-----------------------------|----------------------------|
| New Year's Day              | Rotates     | Long TY   | After                | Buy TY New Years +1, Sell TY New Years +4                              | New Year + 1                | New Year + 4               |
| Martin Luther King Day 1    | Monday      | Long TY   | Before               | Buy TY Tuesday before MLK, Sell Friday before MLK                      | MLK - 4                     | MLK - 1                    |
| Martin Luther King Day 2    | Monday      | Short TY  | After                | Sell TY Tuesday after MLK, Buy Friday after MLK                        | MLK + 1                     | MLK + 4                    |
| President's Day             | Monday      | Long TY   | After                | Buy TY Tuesday after Pres. Day, Sell TY Friday after Pres. Day         | Pres. Day + 1               | Pres. Day + 4              |
| Good Friday                 | Friday      | Long TY   | After                | Buy TY Monday after Good Friday, Sell TY Friday after Good Friday      | Good Friday + 1             | Good Friday + 5            |
| Memorial Day                | Monday      | Long TY   | After                | Buy TY Tuesday after Mem. Day, Sell TY Friday after Mem. Day           | Mem. Day + 1                | Mem. Day + 4               |
| Independence Day (July 4th) | Rotates     | Long TY   | After                | Buy TY Independence Day + 2, Sell TY Independence Day + 4              | Ind. Day + 2                | Ind. Day + 4               |
| Labor Day                   | Monday      | Long TY   | Before               | Buy TY Monday before Labor Day, Sell TY Thursday before Labor Day      | Labor Day - 4               | Labor Day - 2              |
| Columbus Day                | Monday      | Short TY  | Before               | Sell TY Monday before Col. Day, Buy TY Thursday before Col. Day        | Colum. Day - 5              | Colum. Day - 2             |
| Thanksgiving                | Thursday    | Long TY   | Before               | Buy TY Friday before Thanksgiving, Sell TY Tuesday before Thanksgiving | Thanksgiving - 4            | Thanksgiving - 2           |
| Christmas                   | Rotates     | Short TY  | Before               | Sell TY Christmas Day - 4, Buy TY Christmas Day - 1                    | Christmas - 4               | Christmas - 1              |

Source: Morgan Stanley Research. Note: Past performance is no guarantee of future results.



# Global Macro Strategy Team

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| Stock Rating Category | Coverage Universe |            | Investment Banking Clients (IBC) |                |                      | Other Material Investment Services Clients (MISC) |                       |
|-----------------------|-------------------|------------|----------------------------------|----------------|----------------------|---------------------------------------------------|-----------------------|
|                       | Count             | % of Total | Count                            | % of Total IBC | % of Rating Category | Count                                             | % of Total Other MISC |
| Overweight/Buy        | 1499              | 41%        | 389                              | 46%            | 26%                  | 702                                               | 41%                   |
| Equal-weight/Hold     | 1618              | 44%        | 375                              | 44%            | 23%                  | 782                                               | 45%                   |
| Not-Rated/Hold        | 4                 | 0%         | 1                                | 0%             | 25%                  | 1                                                 | 0%                    |
| Underweight/Sell      | 577               | 16%        | 88                               | 10%            | 15%                  | 234                                               | 14%                   |
| Total                 | 3,698             |            | 853                              |                |                      | 1719                                              |                       |

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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